

**UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT**

QWAQWA CAMPUS/QWAQWA-KAMPUS

CSIS3744

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

CONTACT NUMBER/KONTAKNOMMER: 051 401 2929

**EXAMINATION/EKSAMEN:
Additional End-of-year/Bykomende Jaareind 2016
PAPER/VRAESTEL 1**

ASSESSOR(S)/ 1. Mnr. W.S.J. Marais
ASSESSOR(E): 2. Mr. A.G. Musa

MODERATOR(S)/
MODERATOR(E): 1. Dr. B.P. Haskins

TIME/TYD: 180 min

MARKS/PUNTE: 100

INSTRUCTIONS: Answer all the questions. Number your questions clearly and correctly. The Afrikaans version of the questions is in italics.

INSTRUKSIES: *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawe van die vrae is kursief gedruk.*

Question / Vraag 1 [20 Marks / Punte]

Offensive Security has sixteen branches spread across four countries with four branches in each country. Each branch is situated in a specific city. The relevant cities, in which the branches are located, and their applicable countries are listed below: /

Offensive Security het sestiën takke versprei oor vier lande met vier takke in elke land. Elke tak is geleë in 'n spesifieke stad. Die betrokke stede, waarin the takke geleë is, en hulle toepaslike lande word hieronder gelys:

- America / Amerika
 - Chicago
 - Los Angeles
 - New York
 - Washington DC*
- England / Engeland
 - Liverpool
 - London / Londen*
 - Manchester
 - Oxford
- Germany / Duitsland
 - Berlin / Berlyn*
 - Frankfurt
 - Hamburg
 - Stuttgart
- South Africa / Suid-Afrika
 - Bloemfontein
 - Cape Town / Kaapstad
 - Durban
 - Johannesburg*

* Capital Cities / Hoofstede

- 1.1 Offensive Security has implemented a tiered physical topology on their WAN.
The capital cities (indicated by *) form the top tier in the hierarchy.

Question / Vraag 2 [15 Marks / Punte]

Consider the following information regarding Offensive Security's WAN and answer the questions that follow thereafter: /

Neem die volgende inligting ten opsigte van Offensive Security se WAN in ag en beantwoord die vrae wat daarop volg:

- Offensive Security makes use of third party network communication. /

Question / Vraag 3 [30 Marks / Punte]

The following is a network diagram of a 1000Base-T client/server LAN that Offensive Security implemented at each branch: /

Die volgende is 'n netwerkdiagram van 'n 1000Base-T kliënt/bediener-LAN wat Offensive Security by elke tak geïmplementeer het:



- 3.5 Offensive Security wants to separate their network by means of VLANs. /
Offensive Security wil hul netwerk skei deur middel van VLAN's.
- 3.5.1 Which connectivity device would the administrator have to reconfigure to implement the VLANs? /
Watter verbindingstoestel sal die administrateur moet verstel om die VLAN's te implementeer? (1)
- 3.5.2 What is the purpose of an access port and a trunk port on a switch respectively? /
Wat is die doel van 'n toegangspoort en 'n stampoort op 'n skakelaar onderskeidelik? (2)
- 3.5.3 Explain the difference between how a switch and hub operates. /
Verduidelik die verskil tussen hoe 'n skakelaar en spil funksioneer. (2)
- 3.6 Offensive Security wishes to replace the current servers with multiple virtual servers, including a mail and webserver, on a single host machine. Explain to them the disadvantages that this might hold. /
Offensive Security wil die huidige bedieners vervang met veelvuldige virtuele bedieners, insluitend 'n e-pos en web bediener, op 'n enkele gasheermasjien. Verduidelik aan hulle die nadele wat dit mag inhou. (4)
- 3.6.1 Which virtual network connection type must the virtual servers use to be hidden from the physical network, but still be able to communicate with it? /
Watter virtuele netwerk-konneksietipe moet deur die virtuele bedieners gebruik word om van die fisiese netwerk verskuil te wees, maar nogsteeds in staat wees om met dit te kan kommunikeer? (1)
- 3.6.2 Specify the default port addresses that the mail and web servers will use for incoming connections. /
Spesifiseer die verstek poort-adresse wat die e-pos en web-bedieners sal gebruik vir inkomende verbindinge. (2)
- 3.7 What protocol is used to transfer images and formatted text within e-mail messages? /
Watter protokol word gebruik om beelde en geformateerde teks binne e-pos boodskappe oor te dra? (1)
- 3.7.1 Describe how this protocol achieves its objective. /
Beskryf hoe hierdie protokol sy doel bereik. (2)
- [30]

Question / Vraag 4 [15 Marks / Punte]

Offensive Security's wireless access points are configured as follows: /
Offensive Security se draadlose toegangspunte is as volg opgestel:

- SSID: OffSecWiFi / *SSID: OffSecWiFi*
- Encryption standard: WPA2 / *Enkripsiestandaard: WPA2*
- DHCP: Enabled / *DHCP: Aangeskakel*
- Wi-Fi Protected Setup: Enabled / *WiFi Beskermd Opstelling: Aangeskakel*
- Supported 802.11 standards: n/ac / *Ondersteunde 802.11 standaarde: n/ac*

- 4.1 Identify and explain all the relevant security risks that are contained in the access point's configuration settings and give a solution for the vulnerabilities. / *Identifiseer en verduidelik al die tersaaklike veiligheidsrisiko's vervat in die toegangspunt se konfigurasie-instellings en gee 'n oplossing vir die kwesbaarhede.* (2)
- 4.1.1 Explain why WPA2 is a better encryption protocol than WEP. / *Verduidelik waarom WPA2 'n beter enkripsie-protokol is as WEP.* (2)
- 4.2 Name and discuss the access method that is implemented by the wireless access points. / *Noem en bespreek die toegangsmetode wat deur die draadlose toegangspunte geïmplementeer word.* (4)
- 4.3 Does the wireless network make use of simplex, half-duplex or full-duplex transmission? / *Maak die draadlose netwerk gebruik van simpleks, half-dupleks of vol-dupleks transmissie?* (1)
- 4.4 Explain how the wireless network can be set up to allow the users to roam between the access points. / *Verduidelik hoe die draadlose netwerk opgestel kan word om die gebruikers toe te laat om rond te beweeg tussen die toegangspunte.* (2)
- 4.5 Explain the difference between passive scanning and active scanning during the association process. / *Veduidelik die verskil tussen pasiewe skandering en aktiewe skandering gedurende die assosiasie proses.* (2)
- 4.6 On what levels of the OSI model do the following devices operate? / *Op watter vlakke van die OSI-model funksioneer die volgende toestelle?*
- 4.6.1 Wi-Fi access point. / *Wi-Fi toegangspunt.* (1)
- 4.6.2 Router. / *Roetebepaler.* (1)
- [15]

Question / Vraag 5 [20 Marks / Punte]

- 5.1 What server is necessary for users to connect to a LAN from a remote location using the Point-to-Point Protocol (PPP)? / *Watter bediener is nodig vir gebruikers om by 'n LAN aan te sluit van 'n afgeleë plek deur die Punt-na-Punt-Protokol (PPP) te gebruik?* (1)
- 5.1.1 Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? / *Waarom is Punt-tot-Punt Protokol (PPP) 'n beter protokol as Serie-Lyn Internet Protokol (SLIP)?* (5)
- 5.1.2 What protocol is used for connecting home computers to an ISP via DSL? / *Watter protokol word gebruik vir die aansluiting van huisrekenaars aan 'n ISP via DSL?* (1)

- 5.1.3 Explain how an ADSL connection is established between a home computer and the ISP. /
Verduidelik hoe 'n ADSL-verbinding geskep word tussen 'n huisrekenaar en die ISP. (5)
- 5.2 Discuss the difference between a site-to-site VPN and a client-to-site VPN. /
Bespreek die verskil tussen 'n "site-to-site" VPN en 'n "client-to-site" VPN. (4)
- 5.3 What form of multiplexing is implemented by a SONET multiplexer? /
Watter vorm van multipleksering word deur 'n SONET-multiplekseerder geïmplementeer? (1)
- 5.3.1 Briefly explain how this form of multiplexing operates. /
Verduidelik kortliks hoe hierdie vorm van multipleksering werk. (3)
- [20]

End of Paper / Einde van Vraestel

